# APL wiring per glomerulus: the connectome numbers the model uses

> **QUESTION**
> How much APL wiring does each antennal-lobe glomerulus get in the calyx, measured per unit of its input to the KCs, and is that pattern the same in hemibrain and male CNS?

**INPUTS**

- `PN_calyx_weights_APL_vs_KC.csv`: per PN, its calyx synapses onto KCs and to and from APL (Report 01)
- `odor_weighted/KC_APL_synapses_on_claws_by_glomerulus.csv`: KC↔APL synapses that sit on a claw, with the claw's glomerulus (Report 02)
- `hemibrain_KC_claws_derived.csv.gz`, `malecns_KC_claws_derived.csv.gz`: the claws (Report 02)
- `KC_weights_APL_PN_per_KC.csv`: per KC, its calyx APL synapses (Report 01)

**OUTPUTS**

- `odor_weighted/APL_wiring_by_glomerulus.csv` (one row per dataset × glomerulus)
- the weights $p_g$, $q_g$ and the bouton share $f$ used in Report 05

**SCRIPTS**

- `scripts/apl_block_model.py`, section "1. APL wiring per glomerulus"
- `scripts/reports/r04_glomerulus_wiring.py` (this report)

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Report 05 lets APL's inhibition differ between glomeruli in three of its versions (P, L and M). This report explains where those per-glomerulus numbers come from. Only hemibrain and male CNS are used (Yang, 2026-10-21), with official ROIs only. In the male CNS the two calyces are pooled here, because the per-glomerulus numbers are computed per dataset.

**Table 1. Symbols used in this report.**

| Symbol | What it stands for | Values, units, range |
|---|---|---|
| $g$ | An antennal-lobe glomerulus. Only uniglomerular PN types count (one glomerulus in the type name). | hemibrain names VC3l, VC3m, VC5 are renamed VC3, VC5, VM6 to match the other datasets |
| $\mathrm{PN}(g)$ | The uniglomerular PNs of glomerulus $g$ | |
| $\mathrm{claws}(g)$ | Claws whose main PN belongs to $g$ (Report 02) | |
| $n_g^{\mathrm{PN}\to\mathrm{KC}}$ | PN→KC synapses of $\mathrm{PN}(g)$ in the calyx (onto any KC) | synapse count |
| $n_g^{\mathrm{claw}}$ | PN synapses on $\mathrm{claws}(g)$ (from any PN) | synapse count |
| $\lvert\mathrm{claws}(g)\rvert$ | Number of claws of $g$ | count |
| $p_g$ | Bouton weight: APL→PN synapses on $\mathrm{PN}(g)$ in the calyx ÷ $n_g^{\mathrm{PN}\to\mathrm{KC}}$ | synapses per synapse |
| $q_g$ | Claw weight: APL→KC synapses on $\mathrm{claws}(g)$ ÷ $n_g^{\mathrm{claw}}$ | synapses per synapse |

| Symbol | What it stands for | Values, units, range |
|---|---|---|
| $f$ | Boutons' share of APL's calyx output: APL→PN ÷ (APL→PN + APL→KC), all calyx synapses | per dataset |
| $\rho_s$ | Spearman rank correlation, here between datasets over glomeruli | −1 to 1 |

## 1 The units: PN boutons, claws and microglomeruli

In the calyx, each PN axon forms **boutons**: swellings that carry its output synapses. Several KC dendritic **claws** wrap around one bouton, and the bouton with its claws is called a **microglomerulus**. APL makes synapses in both directions with both partners: APL→PN and PN→APL on the boutons, and APL→KC and KC→APL on the claws (Figure 1). Report 02 explains how boutons and claws are found from synapse positions (clusters of sites within 2 µm). A KC↔APL synapse counts as **on a claw** if its KC-side site is within 2 µm of one of that KC's claw PN synapses, the same rule that defines claws. Every claw belongs to one glomerulus: that of its main PN, the PN that makes most of its PN synapses.

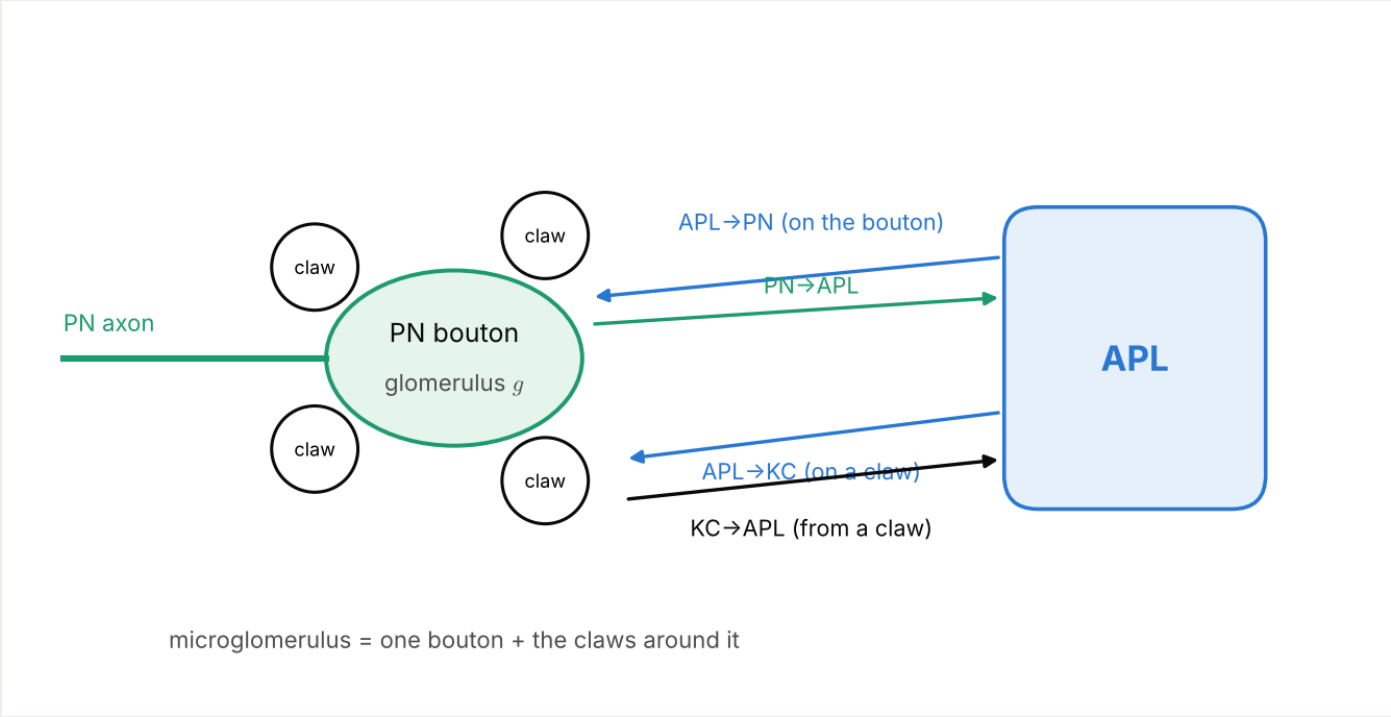



**Figure 1. The four kinds of APL synapse at a microglomerulus.** One PN bouton (green, belonging to glomerulus $g$) with KC claws around it, forming a microglomerulus. APL (blue) makes synapses in four directions there. APL→PN and PN→APL are on the bouton; APL→KC and KC→APL are on the claws. Arrows point from the presynaptic to the postsynaptic cell. The drawing is schematic and not to scale.

## 2 Six per-glomerulus numbers

Raw synapse counts mostly reflect how big a glomerulus is: how many PNs it has and how many claws. To compare glomeruli, each count is divided by a measure of the glomerulus's input to the KCs. For glomerulus $g$:

$$p_g = \frac{\#\,\mathrm{APL}\to\mathrm{PN}\text{ synapses on PN}(g)}{n_g^{\mathrm{PN}\to\mathrm{KC}}}, \qquad p_g^{\mathrm{in}} = \frac{\#\,\mathrm{PN}\to\mathrm{APL}\text{ synapses from PN}(g)}{n_g^{\mathrm{PN}\to\mathrm{KC}}} \tag{1}$$

all synapses in the calyx; $n_g^{\mathrm{PN}\to\mathrm{KC}}$ = PN→KC synapses of the same PNs

$$q_g = \frac{\#\,\text{APL}\to\text{KC on claws}(g)}{n_g^{\text{claw}}}, \qquad q_g^{\text{in}} = \frac{\#\,\text{KC}\to\text{APL on claws}(g)}{n_g^{\text{claw}}} \tag{2}$$

$n_g^{\text{claw}}$ = PN synapses on those claws

Two more divide the claw counts by the number of claws instead ($c_g^{\text{out}}$ and $c_g^{\text{in}}$, APL→KC and KC→APL synapses per claw). The model uses only $p_g$ (version P, and M) and $q_g$ (version L). $p_g$ measures how much APL inhibition a glomerulus's boutons receive for each synapse they pass on to KCs. $q_g$ measures how much APL inhibition its claws receive for each PN synapse they get.

**Table 2. Totals over uniglomerular glomeruli.** Calyx synapses only; male CNS includes both calyces.

| dataset | glomeruli | PNs | PN→KC synapses | APL→PN | PN→APL | claws | claw PN synapses | APL→KC on claws | KC→APL on claws |
|---|---|---|---|---|---|---|---|---|---|
| hemibrain | 56 | 115 | 173,740 | 2,539 | 2,561 | 13,810 | 173,815 | 11,827 | 4,716 |
| male CNS | 57 | 234 | 332,393 | 3,621 | 4,447 | 25,148 | 332,342 | 12,464 | 6,603 |

Small glomeruli give noisy ratios, so the comparisons below use the 50 glomeruli with at least 40 claws and at least 500 PN→KC synapses in both datasets. The model itself uses every glomerulus that has a value.

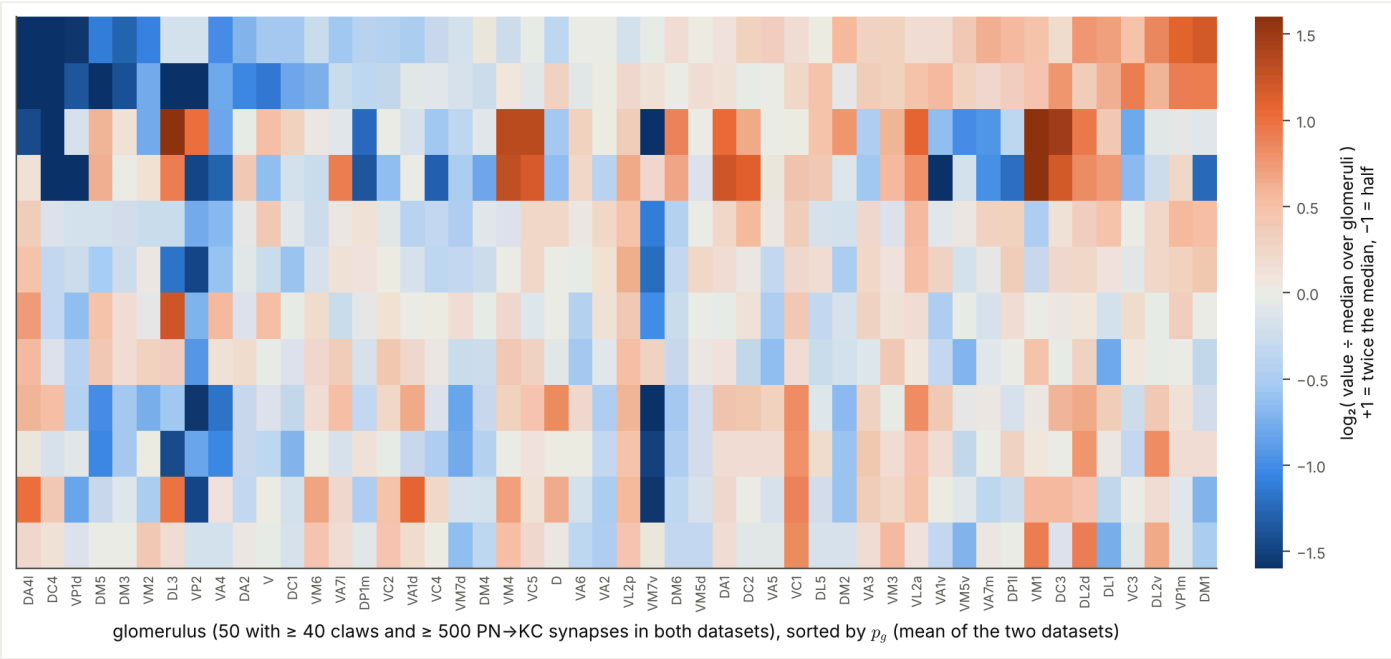



**Figure 2. APL wiring per glomerulus, relative to the median.** Each row is one of the six numbers in one dataset, and each column a glomerulus. Color is the value divided by that row's median over glomeruli, on a $\log_2$ scale: orange above the median, blue below, grey-white at the median. The scale is clipped at ±1.6, about 3× above or below, and a value of 0 (hemibrain DC4 has no APL→PN synapse) takes the darkest blue. Columns are sorted by the bouton weight $p_g$. The top two rows ($p_g$) show the clearest glomerulus-to-glomerulus differences, in the same order in both datasets.

## 3 Which numbers vary, and are they conserved?

**Table 3. Spread across glomeruli and agreement between datasets (50 glomeruli).** The spread is the 10th and 90th percentile over glomeruli divided by the median: "0.5–1.6" means the middle 80% of glomeruli lie between half and 1.6 times the median. The last column is the rank correlation of the glomerulus values between the two datasets.

| number | symbol | hemibrain: 10th–90th percentile (× median) | male CNS: 10th–90th percentile (× median) | Spearman $\rho_s$, hemibrain vs male CNS |
|---|---|---|---|---|

| number | symbol | hemibrain: 10th–90th percentile (× median) | male CNS: 10th–90th percentile (× median) | Spearman $\rho_s$, hemibrain vs male CNS |
|---|---|---|---|---|
| APL→PN per PN→KC | $p_g$ | 0.48–1.53 | 0.37–1.51 | 0.83 |
| PN→APL per PN→KC | $p_g^{\mathrm{in}}$ | 0.52–2.13 | 0.40–2.17 | 0.69 |
| APL→KC on claws per claw PN synapse | $q_g$ | 0.74–1.38 | 0.69–1.25 | 0.75 |
| KC→APL on claws per claw PN synapse | $q_g^{\mathrm{in}}$ | 0.74–1.33 | 0.76–1.32 | 0.53 |
| APL→KC per claw | $c_g^{\mathrm{out}}$ | 0.59–1.52 | 0.58–1.35 | 0.69 |
| KC→APL per claw | $c_g^{\mathrm{in}}$ | 0.70–1.61 | 0.77–1.45 | 0.63 |

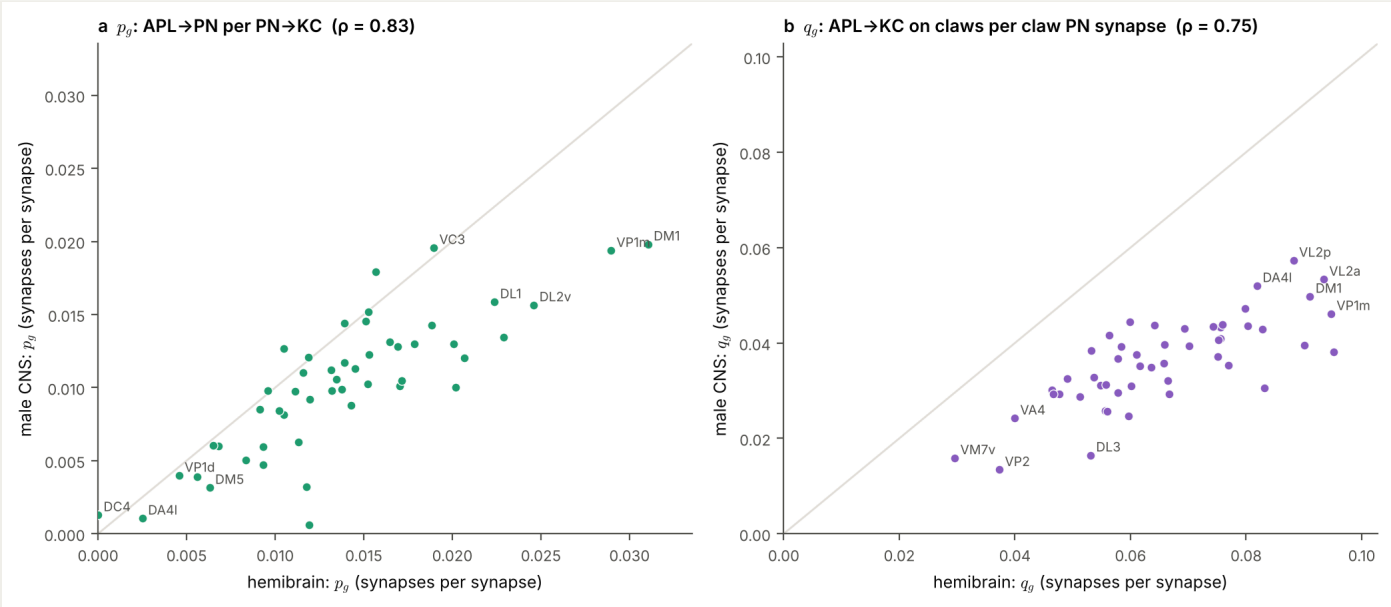



**Figure 3. The two weights the model uses, hemibrain against male CNS.** Each point is one of the 50 glomeruli. Its value in hemibrain is on the horizontal axis, in male CNS on the vertical axis, and the grey line marks equal values. **a**, Bouton weight $p_g$: APL→PN synapses per PN→KC synapse. **b**, Claw weight $q_g$: APL→KC synapses on the claws per claw PN synapse. The labelled glomeruli are the four lowest and five highest (mean of the two datasets, each relative to its median). ρ is the Spearman correlation over glomeruli.

- **Bouton weight** $p_g$ varies about 3–4-fold across glomeruli (10th to 90th percentile), in a similar order in both datasets ($\rho_s$ = 0.83). Lowest: DC4, DA4l, VP1d, DM5, DM3. Highest: DL1, VC3, DL2v, VP1m, DM1.
- **Claw weight** $q_g$ varies only about 2-fold ($\rho_s$ = 0.75).
- The other four numbers agree less between the datasets ($\rho_s$ 0.53–0.69). KC→APL on claws per claw PN synapse agrees least.

## 4 How much of APL's calyx output goes to boutons

Versions M0 and M of Report 05 put only part of the inhibition on the boutons, the same share as APL's synapses:

$$f = \frac{\sum_{\mathrm{PNs}} \#\,\mathrm{APL}{\to}\mathrm{PN}}{\sum_{\mathrm{PNs}} \#\,\mathrm{APL}{\to}\mathrm{PN} + \sum_{\mathrm{KCs}} \#\,\mathrm{APL}{\to}\mathrm{KC}} \tag{3}$$

all calyx synapses of the dataset, not only those on claws or uniglomerular PNs

**Table 4. The bouton share $f$.** About one in ten of APL's calyx output synapses is onto a PN; the rest are onto KCs.

| dataset | APL→PN in the calyx | APL→KC in the calyx | $f$ |
|---|---|---|---|
| hemibrain | 2,738 | 25,609 | 0.097 |
| male CNS | 3,818 | 30,407 | 0.112 |

**Caveats**

Synapse counts stand in for synaptic strength. Nothing here measures how strong a single APL→PN or APL→KC synapse is.

Male CNS official calyx ROIs miss part of the posterior calyx, so its calyx counts are lower than hemibrain's (Report 02). The ratios are less affected, because numerator and denominator lose synapses together.

Glomeruli missing from the response tables still have weights, but they carry no odor input in the model.